

PASSANGER PREFERENCES FOR DOMESTIC AIRLINES OF INDIA

Objective: To analyze passenger preferences for Indian domestic airlines using categorical survey data and perform a right-tailed proportion z-test to verify whether IndiGo is significantly more preferred than other airlines.

Dataset Summary and Statistical Summary (After cleaning)

QUESTION	NO. OF CATEGORIES	MOST CHOSEN AIRLINE	MEAN	MEDIAN	STANDARD DEVIATION	VARIANCE
Most Often Travelled	3	Indigo	17	12	15.3	229
Most Affordable	4	Akasa	12.75	12.5	9.88	97.58
Most Punctual	5	Indigo	10.2	7	12.64	159.7
Most Comfortable	5	Indigo	10.2	9	11.26	126.7
Best Customer Service	6	Indigo	8.5	5.5	9.73	94.7

HYPOTHESIS AND Z-TEST

Hypothesis:
The proportion of passengers who prefer IndiGo is greater than 33.33%.

H1 : $p > 0.3333$

Test Type: One-sample Right-tailed Z-Test (Parametric)

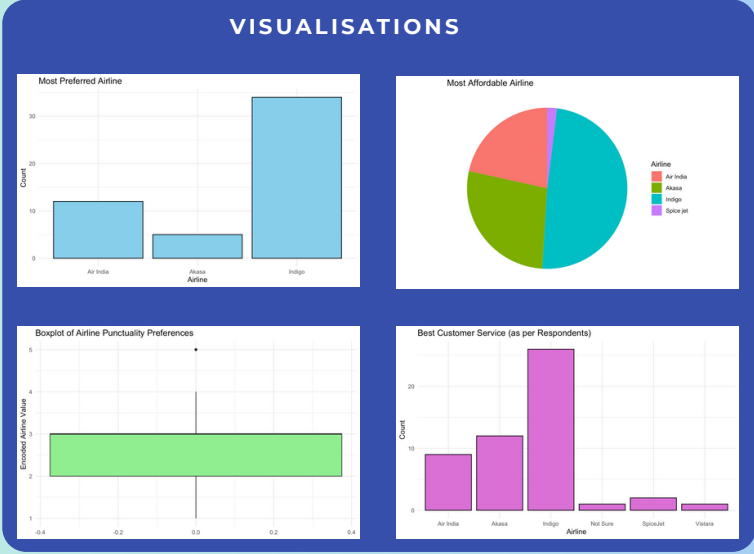
Sample Size (n): 51 IndiGo Count (x): 34

Sample Proportion ($\hat{p}$): 0.6667 p_0 : 0.3333

Z-Statistic: 5.05

p-value: 0.0000

Conclusion:
Since $p < 0.05$, our hypothesis is supported — IndiGo is significantly more preferred than other domestic airlines.



INSIGHTS

- IndiGo dominates across most preference categories (travel frequency, punctuality, service).
- Akasa is seen as the most affordable option.
- High SD in “Most Travelled” → IndiGo’s dominance.
- Z-test confirms a statistically significant preference for IndiGo ($p < 0.05$).

CONCLUSION

The analysis of 51 survey responses shows that IndiGo is statistically the most preferred domestic airline among respondents, while Akasa leads in affordability. The right-tailed z-test confirms that IndiGo’s preference proportion is significantly higher than expected under equal preference ($p < 0.05$).